# 1 Introduction

In classical cryptography, the existence of one-way functions (OWFs) is the minimal assumption [IL89]: OWFs are known to be existentially equivalent to a wide range of cryptographic primitives, including pseudorandom generators (PRGs) [HILL99], pseudorandom functions (PRFs) [GGM86], secret-key encryption (SKE) [GM84], message authentication codes (MACs) [GGM84], digital signatures [Rom90], and commitment schemes [Nao90]. Moreover, nearly all cryptographic primitives (including public-key encryption (PKE) and multiparty computation) imply OWFs.

In contrast, in quantum cryptography, OWFs would not necessarily be the minimal assumption [Kre21, MY22, AQY22]. Several quantum analogues of OWFs, PRGs, and PRFs have been introduced, such as pseudorandom unitaries (PRUs) [JLS18], pseudorandom state generators (PRSGs) [JLS18], one-way state generators (OWSGs) [MY22], one-way puzzles (OWPuzzs) [KT24], and EFI pairs [BCQ23]. They could be weaker than OWFs [Kre21, KQST23, LMW24, KQT24], yet they support a variety of important applications including private-key quantum money [JLS18], SKE [AQY22], MACs [AQY22], digital signatures [MY22], commitments [MY22, AQY22, Yan22], and multiparty computation [MY22, AQY22, BCKM21, GLSV21].

In particular, one-way puzzles (OWPuzzs) introduced by Khurana and Tomer [KT24] are a natural quantum analogue of OWFs, and one of the most fundamental quantum cryptographic primitives. A OWPuzz is a pair $(\mathsf{Samp}, \mathsf{Vrfy})$ of two algorithms. $\mathsf{Samp}$ is a quantum polynomial-time (QPT) algorithm that, on input the security parameter $1^n$, outputs two classical bit strings, $\mathsf{ans}$ and $\mathsf{puzz}$. $\mathsf{Vrfy}$ is an unbounded algorithm that, on input $(\mathsf{puzz}, \mathsf{ans}')$, outputs $\top/\bot$. Correctness requires that $\mathsf{Vrfy}$ accepts $(\mathsf{puzz}, \mathsf{ans}) \leftarrow \mathsf{Samp}(1^n)$ with high probability. Security requires that no QPT adversary that receives $\mathsf{puzz}$ can output $\mathsf{ans}'$ such that $(\mathsf{puzz}, \mathsf{ans}')$ is accepted by $\mathsf{Vrfy}$ with high probability.

The existence of OWPuzzs implies $\mathbf{PP} \neq \mathbf{BQP}$ [CGG$^+$23]. On the other hand, in a recent breakthrough paper by Khurana and Tomer [KT25], OWPuzzs were constructed from $\mathbf{PP} \neq \mathbf{BQP}$ in conjunction with an additional standard assumption in the field of quantum advantage [AA11, BMS16], which we refer to as the "quantum advantage assumption" for simplicity[1]. Although the quantum advantage assumption has been intensively investigated for over a decade, and some partial progress has been made [BFNV19, Mov23], it remains unproven. Moreover, the assumption was a newly introduced one, devised specifically for this purpose, and has not been explored in other areas. As a result, the following question posed by [KT25] remains one of the most important open problems in quantum cryptography.

*Can OWPuzzs be constructed solely from* $\mathbf{PP} \neq \mathbf{BQP}$*?*

This question serves as a quantum analogue of the long-standing open problem in classical cryptography [DH76]: Can OWFs be constructed from $\mathbf{P} \neq \mathbf{NP}$ (or $\mathbf{NP} \not\subseteq \mathbf{BPP}$)? In the classical setting, OWFs and $\mathbf{P} \neq \mathbf{NP}$ (or $\mathbf{NP} \not\subseteq \mathbf{BPP}$) have been extensively studied through the lens of learning hardness [Val84, PV88, IL90, BFKL94, Reg05, NY15, HN23]. Impagliazzo and Levin [IL90] first constructed OWFs based on a specific type of learning hardness, and subsequent works [BFKL94, NR06, NY15, HN23] extended this approach to more standard and general learning models. In particular, OWFs can be constructed from the average-case hardness of proper PAC learning [Val84, BFKL94], while $\mathbf{NP} \not\subseteq \mathbf{BPP}$ implies the worst-case hardness of proper PAC learning [PV88]. These characterizations suggest a promising pathway toward resolving the open problem of constructing OWFs from $\mathbf{NP} \not\subseteq \mathbf{BPP}$, namely, establishing the worst-case to average-case hardness reduction within proper PAC learning! Learning theory is closely connected to meta-complexity [Ko91, HN22, GKLO22, GK23], which is another compelling direction for

[1]This is so-called "anti-concentration" and "average-case $\#\mathbf{P}$-hardness". For the definition, see Assumption 2.10.